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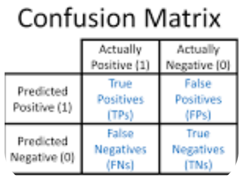
A confusion matrix is **a table that evaluates a classification model's performance by comparing its predicted labels against the actual ground truth labels**. It clearly breaks down correct guesses and specific types of errors, making it easy to see where an algorithm gets confused between different classes. IBM +1


Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

The Four Outcomes of a Binary Confusion Matrix

For a simple yes/no (binary) problem, the table has four main boxes: Oracle Blogs +1

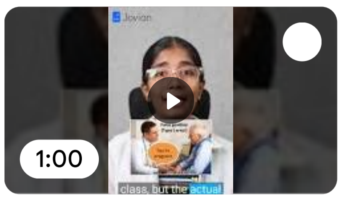
- **True Positive (TP)**: The model predicts yes, and the actual answer is yes.
- **True Negative (TN)**: The model predicts no, and the actual answer is no.
- **False Positive (FP)**: The model predicts yes, but the actual answer is no (Type I error / false alarm).
- **False Negative (FN)**: The model predicts no, but the actual answer is yes (Type II error / missed case). GeeksforGeeks +3

Key Metrics Derived from the Matrix

You can use these four values to calculate important scores: YouTube · ethicalPap_ +1

- **Accuracy**: Total correct predictions divided by all predictions (TP + TN) / (TP + TN + FP + FN).
- **Precision**: True positives divided by all predicted positives TP / (TP + FP). Measures exactness.
- **Recall (Sensitivity)**: True positives divided by all actual positives TP / (TP + FN). Measures completeness.
- **F1-Score**: The balance point between precision and recall. GeeksforGeeks +1

Watch this short guide to see a visual breakdown of how a confusion matrix works:



What is Confusion Matrix in Machine Learning?

12K views · 3 years ago
YouTube · Jovian

If you'd like, let me know:

- Are you working on a **binary** or **multi-class** classification problem?
- Do you need help calculating a specific metric like **precision** or **recall** in Python or another tool?

AI responses may include mistakes. Learn more

IBM

What is a confusion matrix? - IBM

The confusion matrix helps assess classification model performance in machine learning by

Ask anything

Show all

Since I am dyslexic. It also confuses me.

It is completely normal to find it confusing. The name itself even says "confusion," and the layout can look like a jumble of words that bleed together.

To make it much easier to read, let us completely strip away the blocks of text and look at it as a simple **visual checklist** using colors and direct opposites.

🧭 The Two Questions to Ask

Whenever you look at a box in the matrix, you only need to look at two words:

1. **"Did the model get it right?"** → **True** (Yes) or **False** (No).
2. **"What did the model guess?"** → **Positive** (Yes) or **Negative** (No).

🎨 Color-Coded Breakdown

- 💚 **True Positive (TP)** = **Right** guess + Guessed **"Yes"**
- 💚 **True Negative (TN)** = **Right** guess + Guessed **"No"**
- ❤️ **False Positive (FP)** = **Wrong** guess + Guessed **"Yes"** (It shouted "yes" by mistake)
- ❤️ **False Negative (FN)** = **Wrong** guess + Guessed **"No"** (It missed a "yes" completely)

🖼️ Visualizing the Grid Layout

Most matrices are structured like a simple cross. Think of it like this:

	Actual: YES	**Actual: NO**
Predicted: YES	💚 **TP** (Correct "Yes")	❤️ **FP** (False Alarm)
Predicted: NO	❤️ **FN** (Missed Case)	💚 **TN** (Correct "No")

- The **diagonal from top-left to bottom-right** is always the **Good Zone** (Green).
- The **other diagonal** is always the **Error Zone** (Red).

To help make this stick, would you like to try a **quick, real-world example** (like a medical test or a spam email filter) using this color system?

Ask anything

